

Sri Lanka Institute of Information Technology



IT3010 – Network and Distributed Management

Lab Report 02

Installing and Configuring DHCP

Student Name:	Rajapaksha R P M
Student ID:	IT23556102
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Lab Number:	Lab 02

1.Introduction

DHCP is a network protocol used in UDP/IP networks to automatically provide devices with IP addresses and other essential network settings. A DHCP server allocates IP configuration parameters dynamically, allowing devices to join and communicate within a network without requiring manual setup.

In this practical, a DHCP server was configured on a CentOS 9 virtual machine. A Fedora 32 virtual machine was used as the client to confirm that IP addresses were automatically assigned by the server. Both virtual machines were connected using the VMnet2 virtual network within VMware Workstation to enable communication between them.

2.Objectives

- Disable the built-in VMware DHCP service on VMnet 2
- Install and configure a DHCP server on CentOS 9
- Define an IP address pool and subnet configuration in dhcpd.conf
- Start the DHCP service and enable it to start automatically on reboot
- Verify that the Fedora 32 client receives an IP address automatically from the server
- Understand and document the methods for changing IP configurations in Linux

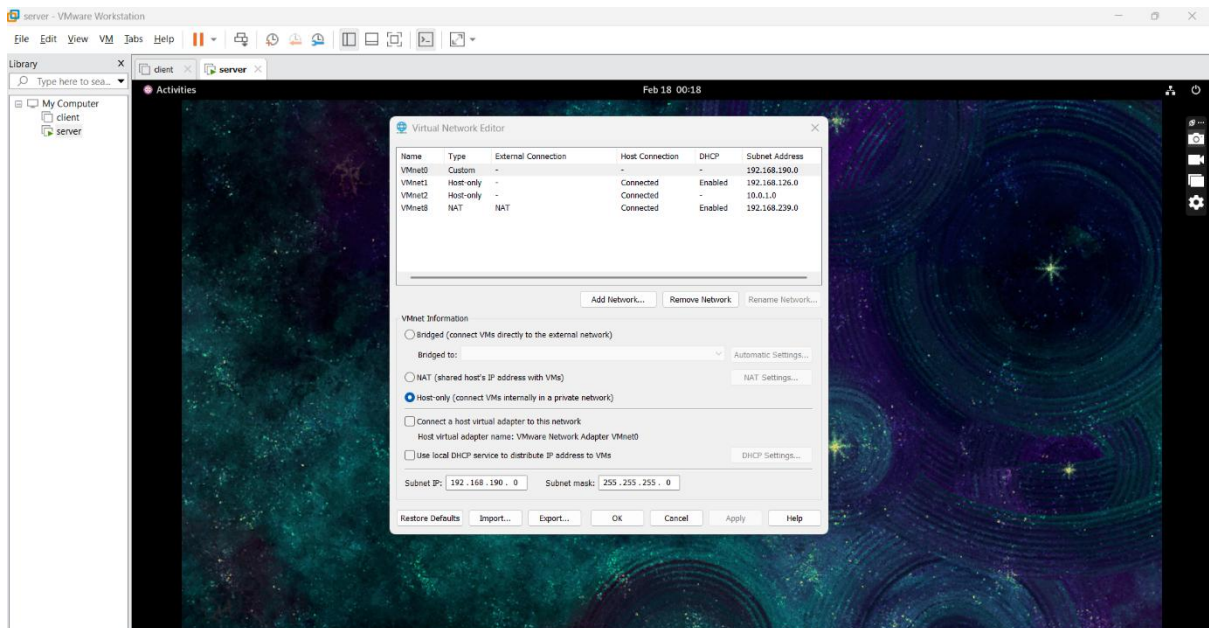
3.Methodology & Steps

The lab was carried out using VMware Workstation with two virtual machines: CentOS 9 (DHCP server) and Fedora 32 (DHCP client), both connected to VMnet 2.

Step 1 – Disabling VMware's Built-in DHCP on VMnet 2

Before configuring a custom DHCP server, the built-in DHCP service provided by VMware on VMnet 2 was disabled to avoid IP address conflicts.

1. Open VMware Workstation
2. Navigate to Edit → Virtual Network Editor
3. Select VMnet 2
4. Uncheck: "Use local DHCP service to distribute IP addresses"
5. Click Apply, then OK

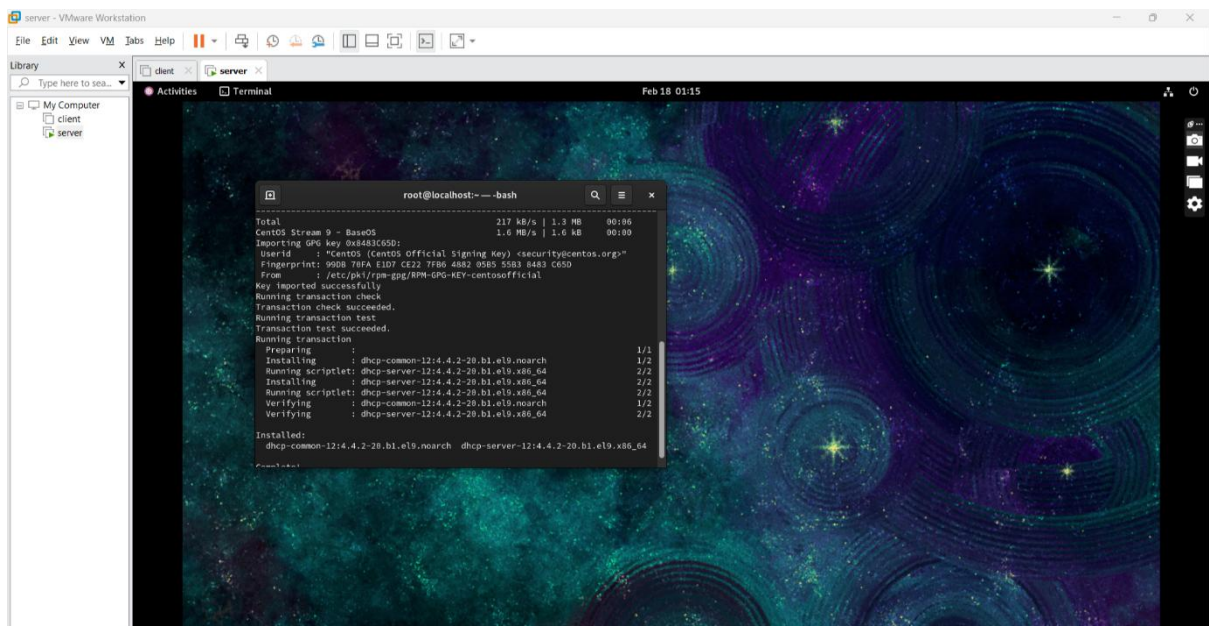


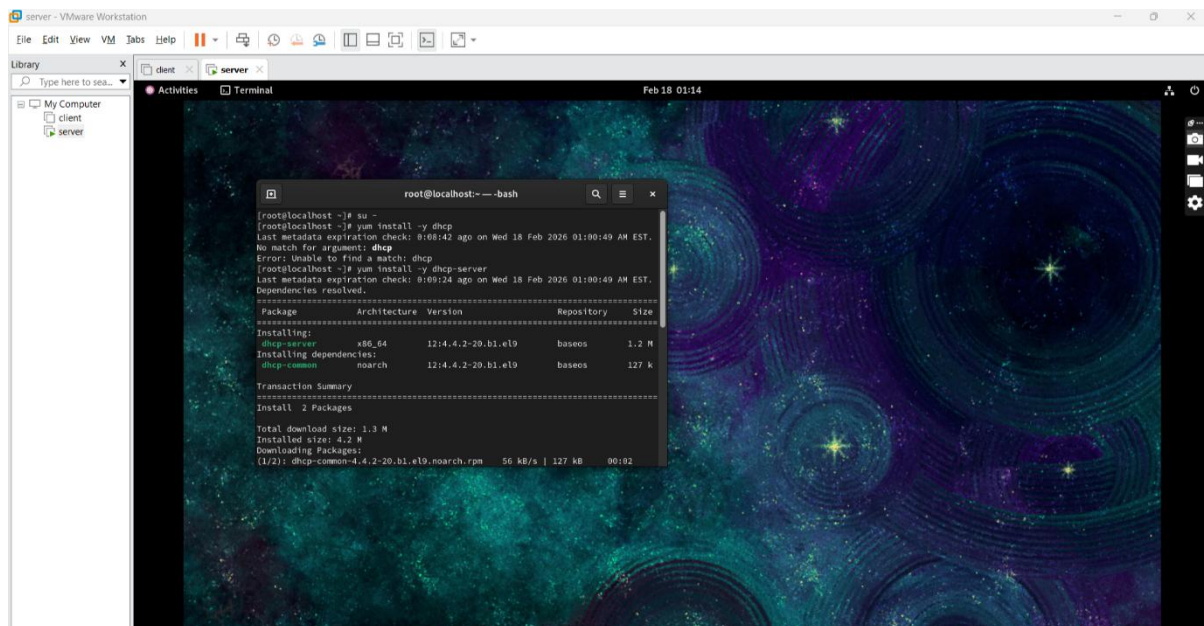
Step 2 – Installing DHCP on CentOS 9

After logging into the CentOS 9 virtual machine as root, the DHCP package was installed using the yum package manager:

```
$ yum install -y dhcp
```

The installation downloaded and installed the dhcp package along with its dependencies. The terminal displayed "Complete!" after a successful install.





Step 3 – Configuring the DHCP Interface

In older versions of CentOS, the DHCP interface was configured by editing the file:
/etc/sysconfig/dhcpd

However, in the current system version, this file is no longer used.

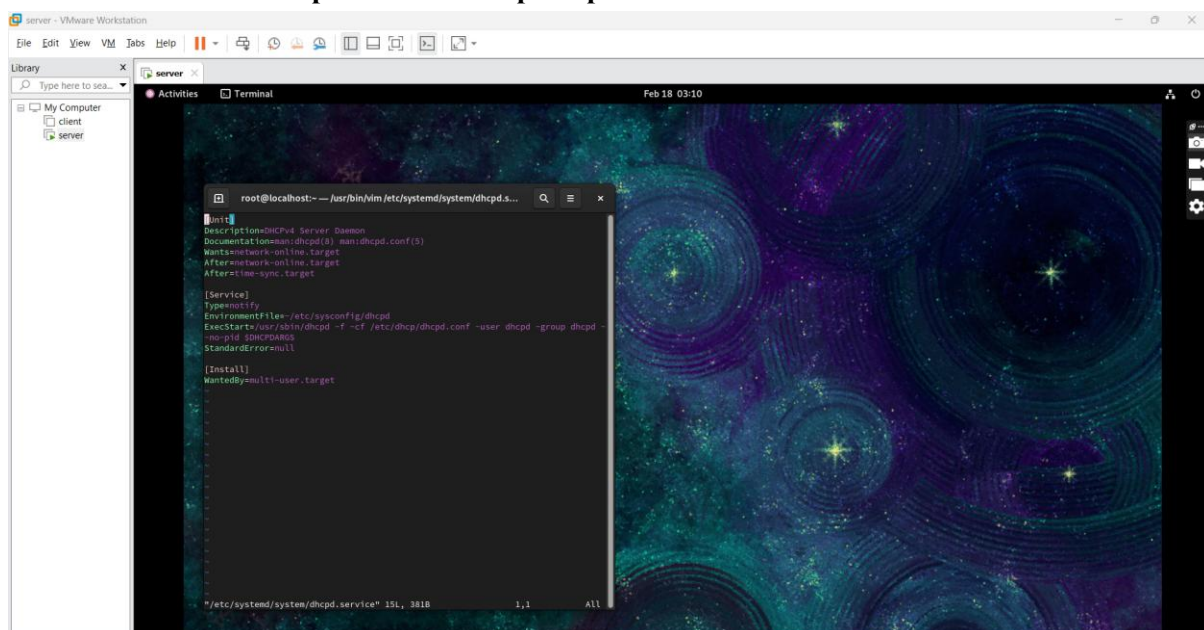
Therefore, the DHCP service file was modified instead.

The following command was used:

vi /etc/systemd/system/dhcpd.service

The ExecStart line was modified to specify the correct interface (ens224):

ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf ens224



Step 4 – Configuring dhcpd.conf

The sample DHCP configuration file was copied to the /etc/dhcp/ directory:

```
cp /usr/share/doc/dhcp-4.2.5/dhcpd.conf.example /etc/dhcp/dhcpd.conf
```

The configuration file was opened and edited:

```
vi /etc/dhcp/dhcpd.conf
```

Key changes made inside the file:

Set domain name and DNS at the top:

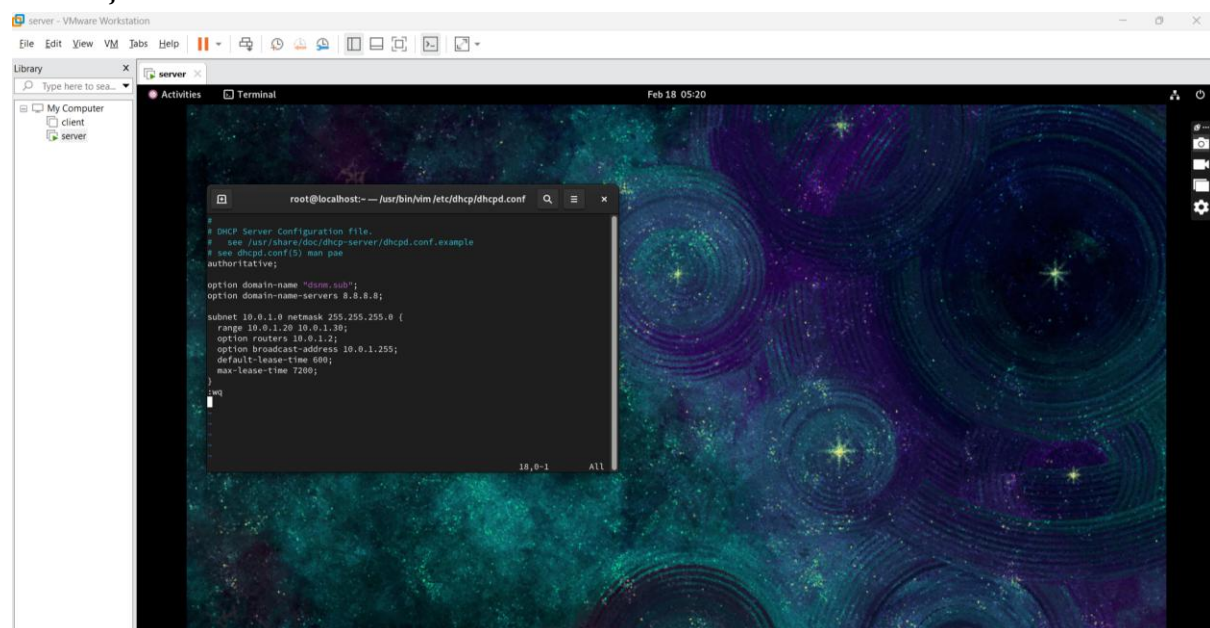
```
option domain-name "dsnm.sub";  
option domain-name-servers server.unixmen.local;  
default-lease-time 600;  
max-lease-time 7200;
```

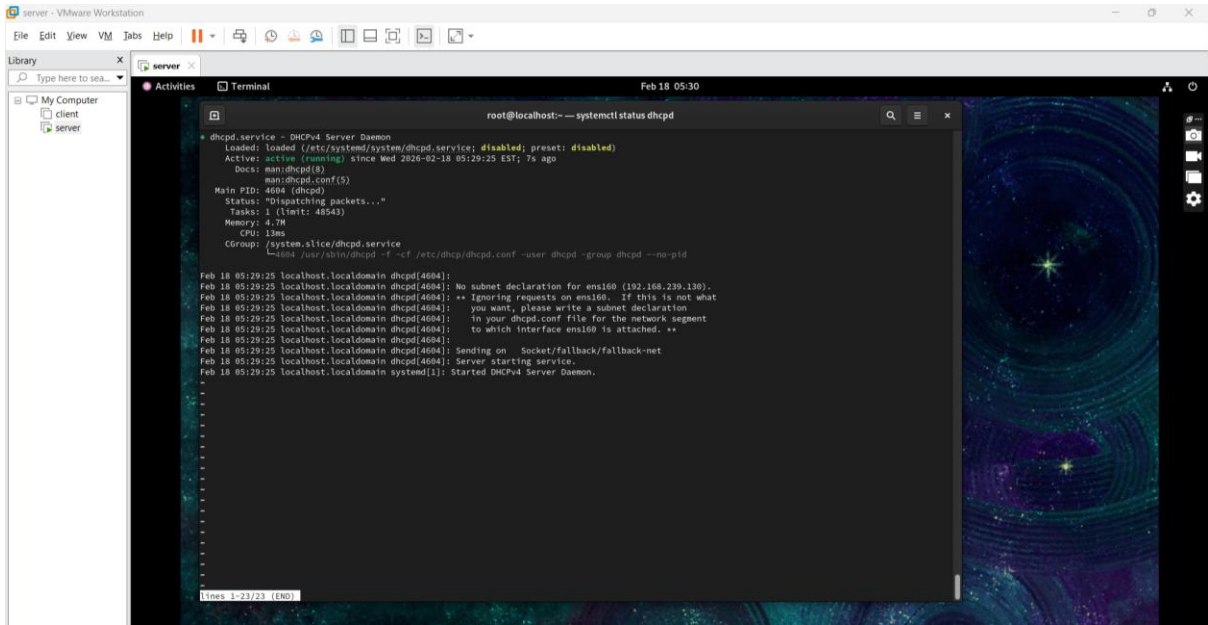
Uncommented the authoritative directive:

```
authoritative;
```

Configured the subnet block:

```
subnet 10.0.1.0 netmask 255.255.255.0 {  
    range 10.0.1.20 10.0.1.30;  
    option domain-name "dsnm.sub";  
    option routers 10.0.1.1;  
    option broadcast-address 10.0.1.255;  
    default-lease-time 600;  
    max-lease-time 7200;  
}
```

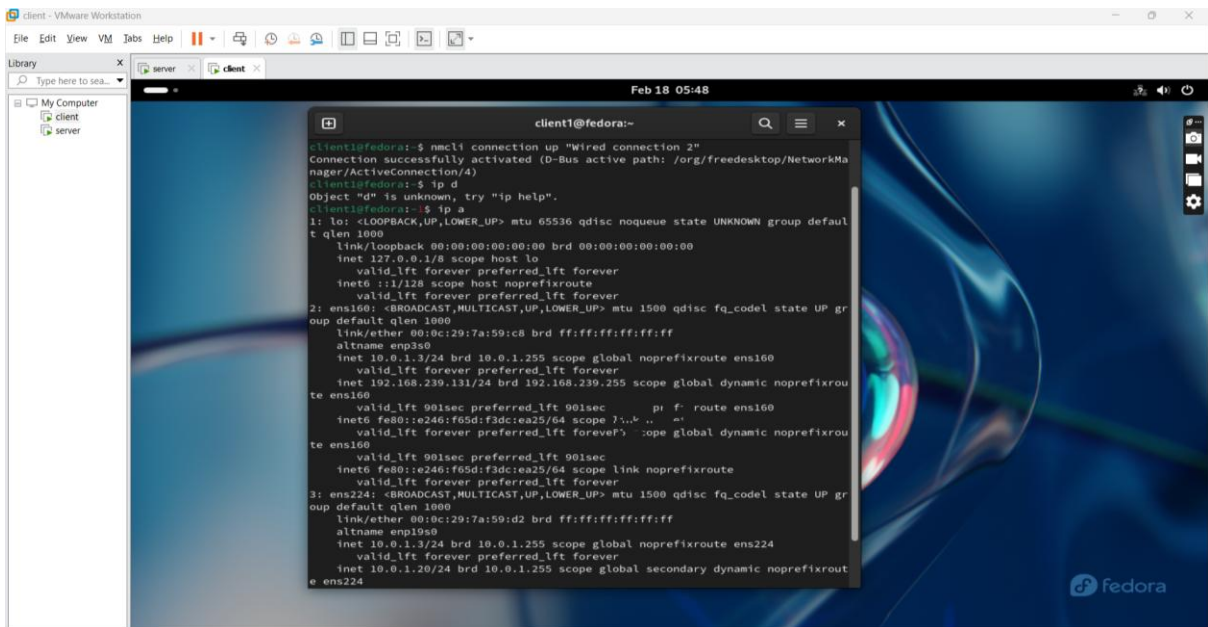


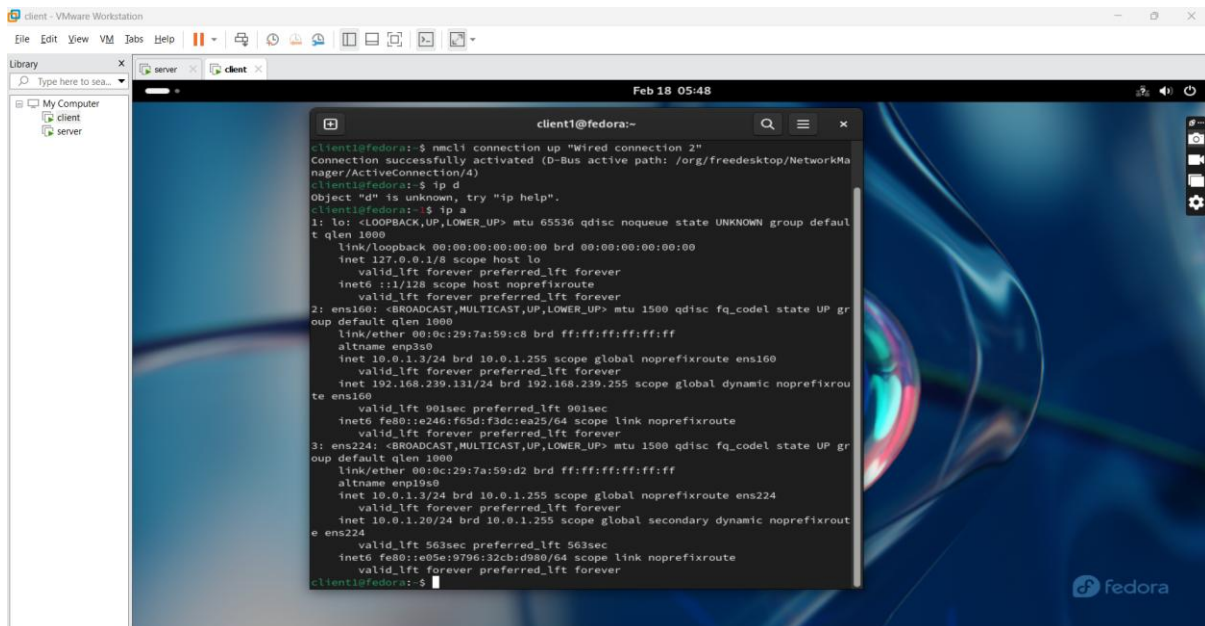


Step 5 – Starting the DHCP Service

The DHCP service was started and enabled to run on every reboot:

```
service dhcpd start
chkconfig dhcpd on
```





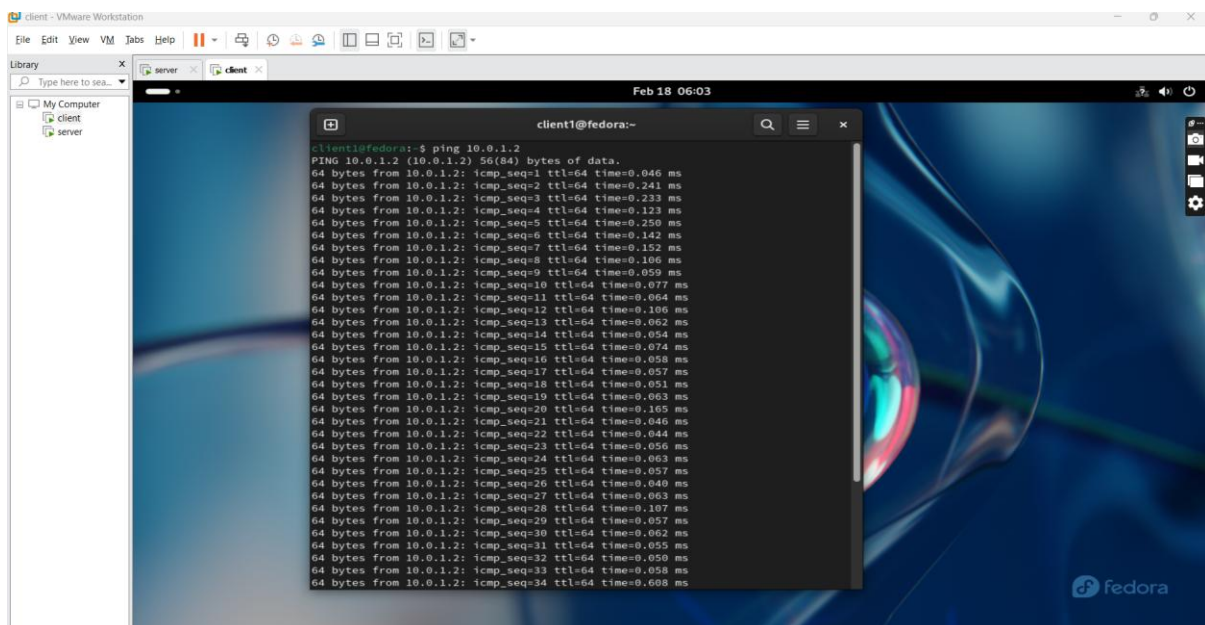
```
client1@fedora:~$ nmcli connection up "Wired connection 2"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/4)
client1@fedora:~$ ip a
client1@fedora:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:7a:59:c8 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet 192.168.239.131/24 brd 192.168.239.255 scope global dynamic noprefixroute ens160
        valid_lft 901sec preferred_lft 901sec
    inet6 fe80::e246:f65d:f3dc:ea25/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:7a:59:d2 brd ff:ff:ff:ff:ff:ff
    altname enp19s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens224
        valid_lft forever preferred_lft forever
    inet 10.0.1.20/24 brd 10.0.1.255 scope global secondary dynamic noprefixroute ens224
        valid_lft 563sec preferred_lft 563sec
    inet6 fe80::e05e:9796:32cb:d980/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
client1@fedora:~$
```

Step 6 – Verifying DHCP Server Status

The status of the DHCP service was confirmed using:

systemctl status dhcpd

The output confirmed the service was active (running) and showed DHCPDISCOVER/DHCPREQUEST log messages from the Fedora client.



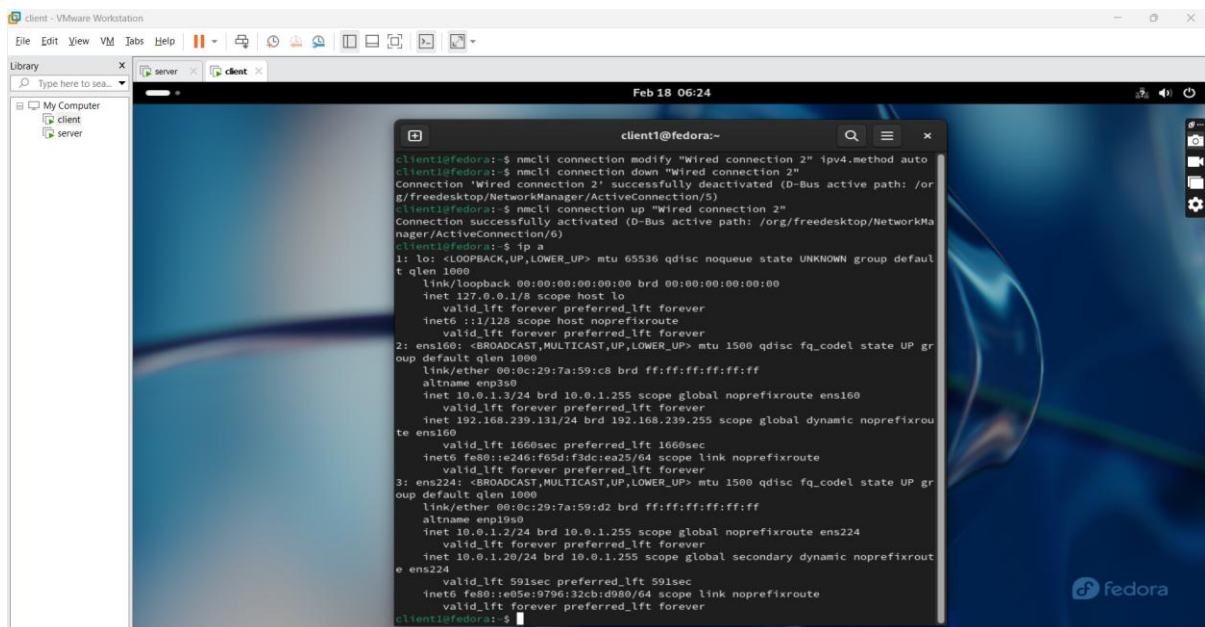
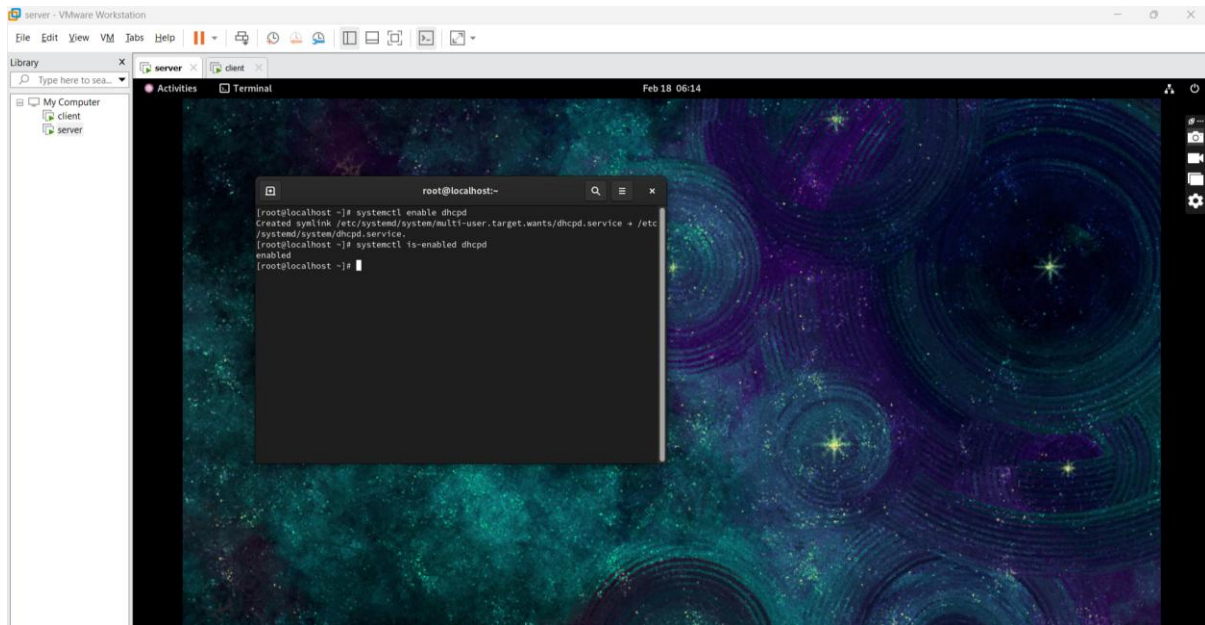
```
client1@fedora:~$ ping 10.0.1.2
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data:
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=0.046 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=0.241 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=0.233 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=0.123 ms
64 bytes from 10.0.1.2: icmp_seq=5 ttl=64 time=0.250 ms
64 bytes from 10.0.1.2: icmp_seq=6 ttl=64 time=0.142 ms
64 bytes from 10.0.1.2: icmp_seq=7 ttl=64 time=0.152 ms
64 bytes from 10.0.1.2: icmp_seq=8 ttl=64 time=0.106 ms
64 bytes from 10.0.1.2: icmp_seq=9 ttl=64 time=0.059 ms
64 bytes from 10.0.1.2: icmp_seq=10 ttl=64 time=0.077 ms
64 bytes from 10.0.1.2: icmp_seq=11 ttl=64 time=0.064 ms
64 bytes from 10.0.1.2: icmp_seq=12 ttl=64 time=0.106 ms
64 bytes from 10.0.1.2: icmp_seq=13 ttl=64 time=0.062 ms
64 bytes from 10.0.1.2: icmp_seq=14 ttl=64 time=0.054 ms
64 bytes from 10.0.1.2: icmp_seq=15 ttl=64 time=0.074 ms
64 bytes from 10.0.1.2: icmp_seq=16 ttl=64 time=0.058 ms
64 bytes from 10.0.1.2: icmp_seq=17 ttl=64 time=0.057 ms
64 bytes from 10.0.1.2: icmp_seq=18 ttl=64 time=0.051 ms
64 bytes from 10.0.1.2: icmp_seq=19 ttl=64 time=0.063 ms
64 bytes from 10.0.1.2: icmp_seq=20 ttl=64 time=0.165 ms
64 bytes from 10.0.1.2: icmp_seq=21 ttl=64 time=0.046 ms
64 bytes from 10.0.1.2: icmp_seq=22 ttl=64 time=0.044 ms
64 bytes from 10.0.1.2: icmp_seq=23 ttl=64 time=0.056 ms
64 bytes from 10.0.1.2: icmp_seq=24 ttl=64 time=0.063 ms
64 bytes from 10.0.1.2: icmp_seq=25 ttl=64 time=0.057 ms
64 bytes from 10.0.1.2: icmp_seq=26 ttl=64 time=0.040 ms
64 bytes from 10.0.1.2: icmp_seq=27 ttl=64 time=0.063 ms
64 bytes from 10.0.1.2: icmp_seq=28 ttl=64 time=0.107 ms
64 bytes from 10.0.1.2: icmp_seq=29 ttl=64 time=0.057 ms
64 bytes from 10.0.1.2: icmp_seq=30 ttl=64 time=0.062 ms
64 bytes from 10.0.1.2: icmp_seq=31 ttl=64 time=0.055 ms
64 bytes from 10.0.1.2: icmp_seq=32 ttl=64 time=0.050 ms
64 bytes from 10.0.1.2: icmp_seq=33 ttl=64 time=0.058 ms
64 bytes from 10.0.1.2: icmp_seq=34 ttl=64 time=0.608 ms
```

Step 7 – Testing on the Fedora Client

On the Fedora 32 VM, the network connection settings were changed to Automatic (DHCP) and saved. The IP address was then verified:

ifconfig

The output confirmed the Fedora machine received an IP address in the range 10.0.1.20 – 10.0.1.30.

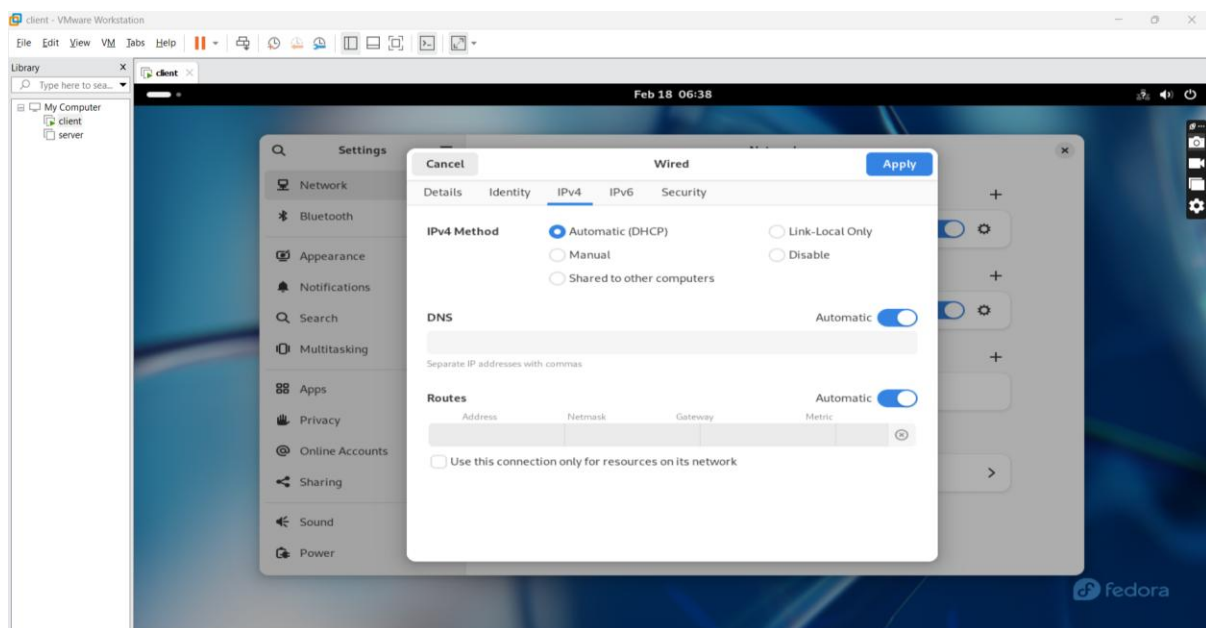


Step 4 – Self-Study – Changing IP Configurations in Linux

There are three main methods that can be used to change IP configurations in a Linux system.

Method 1 – GUI Method

The graphical method involves using the Network Manager interface. The user can navigate to network settings, select the active connection, and manually enter IP address, netmask, gateway, and DNS server details. This method requires a desktop environment and is not suitable for headless servers.

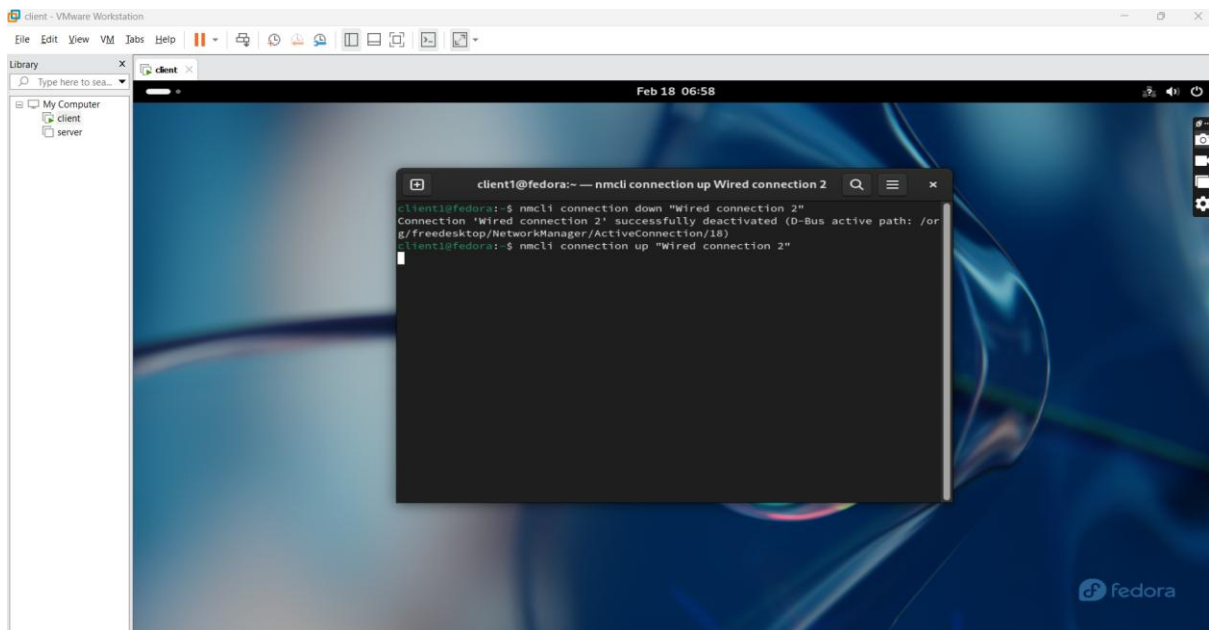


Method 2 – CLI Method Using nmcli / nmtui

This method was used in Lab 01. The nmcli tool manages network connections from the terminal. After editing connection settings, the interface is restarted with:

```
nmcli connection down <connection name>
```

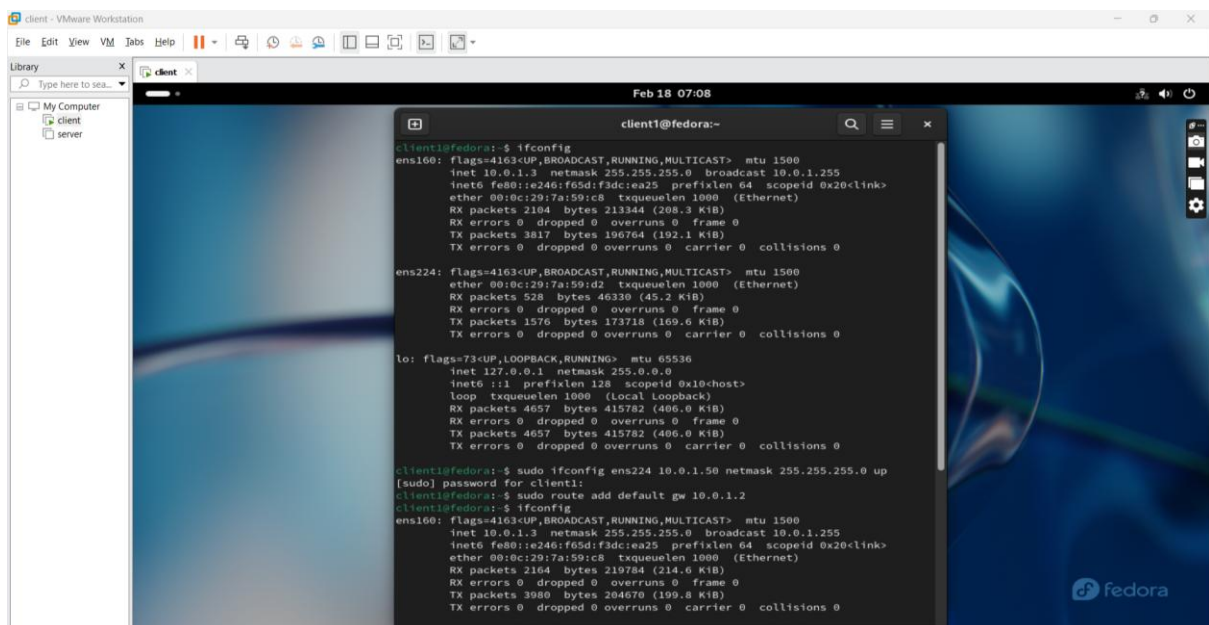
```
nmcli connection up <connection name>
```

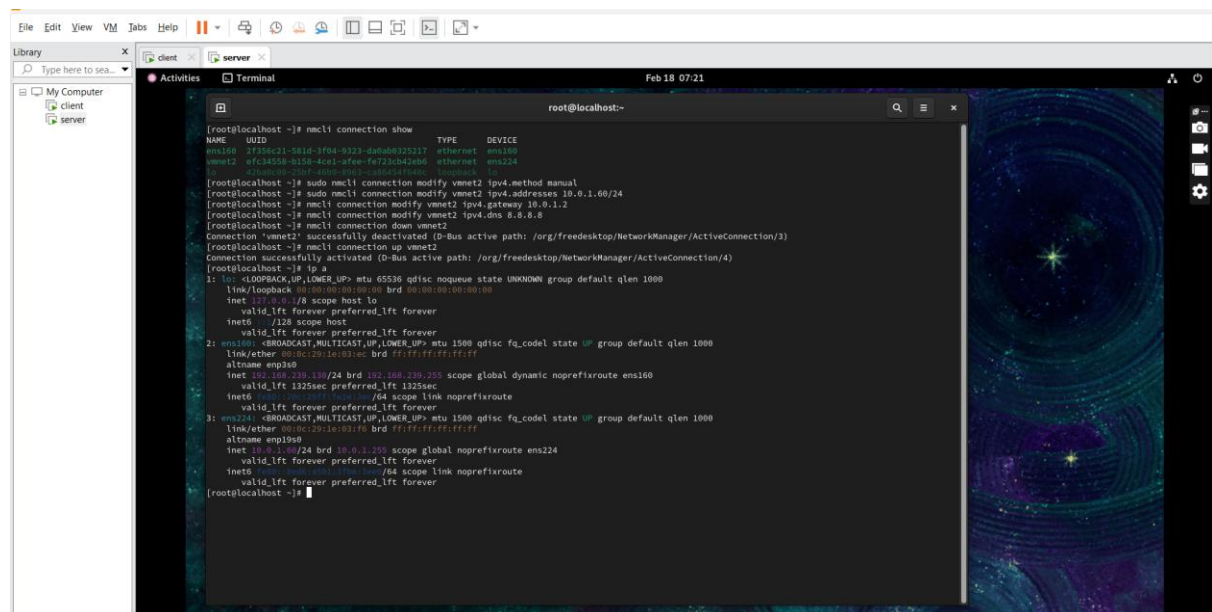


Method 3 – Assigning a Temporary IP Address

A temporary IP can be assigned using `ifconfig`. This is lost when the interface is restarted.

6. Identify the interface: `ifconfig`
7. Assign temporary IP:
`ifconfig <interface name> up <IP Address> netmask <subnetmask>`
8. Add default gateway:
`route add default gw <Gateway IP>`





Step 5 – Results

Task	Result
Disabled VMware DHCP on VMnet 2	Successfully completed
Installed DHCP package on CentOS 9	Installed successfully via yum
Configured dhcpd.conf with subnet 10.0.1.0/24	Configuration saved without errors
Started dhcpd service	Service active (running)
Enabled dhcpd on boot	Enabled via chkconfig
Fedora received IP from DHCP server	IP assigned in range 10.0.1.20–10.0.1.30

Step 6– Discussion

The lab was completed successfully. Disabling VMware's built-in DHCP was essential to prevent IP conflicts. The authoritative; directive ensures the server sends DHCPNAK to clients requesting addresses outside the defined scope. The difference between default-lease-time (600 sec) and max-lease-time (7200 sec) determines how long IP assignments last. Commenting out all unused subnet blocks in dhcpd.conf was necessary to prevent service startup failures.

Conclusion

This practical effectively demonstrated the process of setting up and configuring two virtual machines using CentOS 9 as the server and Fedora 32 LXDE as the client. Both virtual machines were connected through the VMnet2 virtual network, and static IP addresses were manually assigned to enable communication between them. Successful ping results confirmed that the network configuration was correctly implemented. Overall, this lab strengthened understanding of virtualization concepts, virtual networking, and manual IP configuration using command-line tools.